



Q&A WITH
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‘We do not understand how these systems work’

STUART RUSSELL, PROFESSOR OF COMPUTER SCIENCE AT UNIVERSITY OF CALIFORNIA, BERKELEY, ON WHY GUARD RAILS ARE NECESSARY FOR THE DEVELOPMENT OF AI

S **TUART RUSSELL**, Professor of Computer Science at University of California, Berkeley, who has been an AI researcher for 45 years, in an interaction with *BT*, says that while artificial intelligence can be hugely beneficial, it also has the potential to disrupt the world in a bad way if guard rails are not put in place. Edited excerpts:

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Let's talk about the open letter to halt the development of AI systems more powerful than GPT-4 and develop guardrails. What prompted you to sign this letter?

We're calling for a halt on the deployment of large language models (LLMs) that are more powerful than the ones that have already been released. And the reason is simple: we do not understand how these systems work.

So, what is an LLM? It's a computer program that predicts the next word, given a sequence of preceding words. And with that system, you can have a conversation. The way these systems are built, there's a large amount of training data. In the case of GPT-4, we think [it is around] 20-30 trillion words of text—approximately the same amount we have in all the books that the human race has ever written.

And then we start from what you might think of as a blank slate, an enormous circuit with about a trillion parameters or more. And then by the process of doing about a billion or trillion small random permutations to those parameters, the system is gradually improved, as is its ability to predict the next word. The result of that is something that when you converse with it, has the appearance of an intelligent entity.



CHERNOBYL DESTROYED THE NUCLEAR INDUSTRY... WE DON'T WANT TO HAVE [A CHERNOBYL-LIKE INCIDENT FOR AI]